

Algebra Essentials

ANSWER KEY



Evaluating expressions

- 1 If $x = 4$, evaluate $3x + 7$.
 $3(4) + 7 = 12 + 7 = 19$.
 - 2 If $a = 5$ and $b = 2$, evaluate $a^2 - 3b$.
 $5^2 - 3(2) = 25 - 6 = 19$.
 - 3 If $x = -3$, evaluate $2x^2 + 5x$.
 $2(-3)^2 + 5(-3) = 2(9) - 15 = 18 - 15 = 3$.
 - 4 If $p = 6$ and $q = -2$, evaluate $\frac{p - q}{q + 6}$.
 $\frac{6 - (-2)}{-2 + 6} = \frac{8}{4} = 2$.
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Solving linear equations

- 5 Solve $2x + 5 = 17$.
 $2x = 12$, so $x = 6$.
 - 6 Solve $5x - 8 = 27$.
 $5x = 35$, so $x = 7$.
 - 7 Solve $3(x - 4) = 21$.
 $x - 4 = 7$, so $x = 11$.
 - 8 Solve $\frac{x}{4} + 3 = 10$.
 $\frac{x}{4} = 7$, so $x = 28$.
 - 9 Solve $7x + 3 = 4x + 18$.
 $7x - 4x = 18 - 3$, so $3x = 15$, and $x = 5$.
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Expanding and simplifying

- 10 Expand $(x + 3)(x + 5)$.
 $x^2 + (3 + 5)x + 3 \times 5 = x^2 + 8x + 15$.
- 11 Expand $(x - 4)(x + 6)$.
 $x^2 + (-4 + 6)x + (-4)(6) = x^2 + 2x - 24$.

- 12 Expand $(2x + 1)(x - 3)$.
 $2x^2 - 6x + x - 3 = 2x^2 - 5x - 3$.
- 13 Simplify $3(2x - 5) - 2(x - 4)$.
 $6x - 15 - 2x + 8 = 4x - 7$.
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Factoring

- 14 Factor $x^2 + 9x + 20$.
Find two numbers with sum 9 and product 20: 4 and 5. $(x + 4)(x + 5)$.
- 15 Factor $x^2 - 2x - 15$.
Find two numbers with sum -2 and product -15 : -5 and 3 . $(x - 5)(x + 3)$.
- 16 Factor $x^2 - 49$.
Difference of squares: $(x - 7)(x + 7)$.
- 17 Factor $6x + 15$.
Common factor 3: $3(2x + 5)$.
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Solving quadratic equations by factoring

- 18 Solve $x^2 + 5x + 6 = 0$.
Factor: $(x + 2)(x + 3) = 0$, so $x = -2$ or $x = -3$.
- 19 Solve $x^2 - 7x + 12 = 0$.
Factor: $(x - 3)(x - 4) = 0$, so $x = 3$ or $x = 4$.
- 20 Solve $x^2 - 16 = 0$.
Difference of squares: $(x - 4)(x + 4) = 0$, so $x = 4$ or $x = -4$.
- 21 Solve $x^2 + 2x - 15 = 0$.
Factor: $(x + 5)(x - 3) = 0$, so $x = -5$ or $x = 3$.
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Systems of two equations

- 22 Solve the system: $x + y = 10$, $x - y = 4$.
Adding the equations: $2x = 14$, so $x = 7$. Then $y = 10 - 7 = 3$.
- 23 Solve the system: $2x + y = 11$, $x - y = 1$.
Adding the equations: $3x = 12$, so $x = 4$. Then $y = x - 1 = 3$.

- 24** Solve the system: $x = 3y - 1$, $2x + y = 19$.
Substitute: $2(3y - 1) + y = 19 \Rightarrow 6y - 2 + y = 19 \Rightarrow 7y = 21 \Rightarrow y = 3$. Then $x = 3(3) - 1 = 8$.
- 25** Solve the system: $4x - y = 10$, $2x + y = 8$.
Adding the equations: $6x = 18$, so $x = 3$. Then $y = 8 - 2(3) = 2$.
- 26** Solve the system: $5x + 3y = 21$, $2x - 3y = 0$.
Adding the equations: $7x = 21$, so $x = 3$. Then $3y = 2x = 6$, so $y = 2$.
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Laws of exponents

- 27** Simplify $a^5 \times a^3$.
 $a^{5+3} = a^8$.
- 28** Simplify $\frac{x^9}{\sqrt[4]{x^4}}$.
 $x^{9-4} = x^5$.
- 29** Simplify $(y^3)^4$.
 $y^{3 \times 4} = y^{12}$.
- 30** Simplify $2x^3 \times 5x^2$.
 $2 \times 5 \times x^{3+2} = 10x^5$.
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Inequalities

- 31** Solve $3x - 4 > 11$.
 $3x > 15$, so $x > 5$.
- 32** Solve $-2x + 6 \leq 20$.
 $-2x \leq 14$. Dividing by a negative reverses the inequality: $x \geq -7$.
- 33** Solve $5x + 1 \geq 3x + 9$.
 $2x \geq 8$, so $x \geq 4$.
- 34** Solve $4(x - 2) < 2x + 6$.
 $4x - 8 < 2x + 6 \Rightarrow 2x < 14 \Rightarrow x < 7$.
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Substitution into formulas

- 35** The formula for simple interest is $I = Prt$. Find I when $P = 2,000$, $r = 0.04$, $t = 3$.
 $I = 2,000 \times 0.04 \times 3 = 240$.

- 36 The perimeter of a rectangle is $P = 2(l + w)$. Find P when $l = 15$ and $w = 9$.
 $P = 2(15 + 9) = 2(24) = 48$.
- 37 If $v = u + at$, find v when $u = 5$, $a = 3$, $t = 4$.
 $v = 5 + 3(4) = 5 + 12 = 17$.
- 38 The area of a triangle is $A = \frac{1}{2}bh$. Find A when $b = 14$ and $h = 9$.
 $A = \frac{1}{2} \times 14 \times 9 = 63$.
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Translating words into equations

- 39 Twice a number, plus 7, equals 25. Find the number.
 $2x + 7 = 25 \Rightarrow 2x = 18 \Rightarrow x = 9$.
- 40 A number decreased by 8 is equal to three times the number, decreased by 20. Find the number.
 $x - 8 = 3x - 20 \Rightarrow 12 = 2x \Rightarrow x = 6$.
- 41 The sum of three consecutive integers is 84. Find the integers.
Let the integers be n , $n + 1$, $n + 2$: $3n + 3 = 84 \Rightarrow 3n = 81 \Rightarrow n = 27$. The integers are 27, 28, 29.
- 42 Five more than three times a number is the same as 10 less than five times the number. Find the number.
 $3x + 5 = 5x - 10 \Rightarrow 15 = 2x \Rightarrow x = 7.5$.
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Word problems

- 43 A rectangle has length $(x + 3)$ cm and width x cm. This question has two parts.
- a If the area is 40 cm², write an equation in x . Area = $x(x + 3) = 40$, i.e. $x^2 + 3x - 40 = 0$.
- b Solve for x . Factoring: $(x + 8)(x - 5) = 0$. Since length must be positive, $x = 5$.
- 44 A father is 3 times as old as his son. In 12 years, he will be twice as old as his son. Find their current ages.
Let the son's age be x ; the father's is $3x$. In 12 years:
 $3x + 12 = 2(x + 12) \Rightarrow 3x + 12 = 2x + 24 \Rightarrow x = 12$. Son is 12, father is 36.
- 45 Tickets cost 25 riyals for adults and 15 riyals for children. A group of 10 people paid 220 riyals in total. How many adults were in the group?
Let a be the number of adults, so $10 - a$ are children:
 $25a + 15(10 - a) = 220 \Rightarrow 25a + 150 - 15a = 220 \Rightarrow 10a = 70 \Rightarrow a = 7$ adults.
- 46 The sum of two numbers is 45, and their difference is 11. Find the two numbers.
Let the numbers be x and y with $x + y = 45$ and $x - y = 11$. Adding: $2x = 56 \Rightarrow x = 28$, then $y = 17$.
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MCQ — Evaluating expressions

47 If $x = 6$, evaluate $4x - 9$.

- a. 33
- b. 24
- c. -15
- d. 15 ✓

$$4(6) - 9 = 24 - 9 = 15.$$

48 If $a = -2$ and $b = 3$, evaluate $a^2 + 2ab$.

- a. -16
- b. 8
- c. -4
- d. -8 ✓

$$a^2 = 4, 2ab = 2(-2)(3) = -12. 4 + (-12) = -8.$$

49 If $x = 5$ and $y = -1$, evaluate $(x + y)^2 - xy$.

- a. 16
- b. 11
- c. 21 ✓
- d. -5

$$(x + y)^2 = 4^2 = 16; xy = -5. 16 - (-5) = 21.$$

50 If $p = 3$, evaluate $2p^3 - 5p$.

- a. 49
- b. 39 ✓
- c. 33
- d. 69

$$2(27) - 15 = 54 - 15 = 39.$$

MCQ — Solving linear equations

51 Solve $\frac{x}{3} + 4 = 9$.

a. $x = 27$

b. $x = 15$ ✓

c. $x = 5$

d. $x = 39$

$\frac{x}{3} = 5 \Rightarrow x = 15$.

52 Solve $4(x + 2) = 3x + 15$.

a. $x = 1$

b. $x = 23$

c. $x = -7$

d. $x = 7$ ✓

$4x + 8 = 3x + 15 \Rightarrow x = 7$.

53 Solve $0.5x - 3 = 7$.

a. $x = 20$ ✓

b. $x = 8$

c. $x = -20$

d. $x = 14$

$0.5x = 10 \Rightarrow x = 20$.

54 Solve $\frac{2x-1}{3} = 5$.

a. $x = 4$

b. $x = 7$

c. $x = 16$

d. $x = 8$ ✓

$$2x - 1 = 15 \Rightarrow 2x = 16 \Rightarrow x = 8.$$

55 Solve $6 - 2x = 3x - 9$.

a. $x = -3$

b. $x = 3$ ✓

c. $x = 15$

d. $x = 1$

$$15 = 5x \Rightarrow x = 3.$$

MCQ — Expanding and simplifying

56 Expand $(x + 7)^2$.

a. $x^2 + 7x + 49$

b. $x^2 + 14x + 14$

c. $x^2 + 14x + 49$ ✓

d. $x^2 + 49$

$$(x + 7)^2 = x^2 + 2(7)x + 49 = x^2 + 14x + 49.$$

57 Expand $(x - 5)^2$.

a. $x^2 + 10x + 25$

b. $x^2 - 5x + 25$

c. $x^2 - 10x + 25$ ✓

d. $x^2 - 25$

$$(x - 5)^2 = x^2 - 10x + 25.$$

58 Expand $(3x - 2)(3x + 2)$.

a. $9x^2 + 4$

b. $9x^2 - 12x - 4$

c. $9x^2 - 4$ ✓

d. $6x^2 - 4$

$$\text{Difference of squares: } (3x)^2 - 2^2 = 9x^2 - 4.$$

59 Simplify $2(x + 3) - 3(x - 1)$.

a. $5x + 9$

b. $-x + 9$ ✓

c. $x + 9$

d. $-x + 3$

$$2x + 6 - 3x + 3 = -x + 9.$$

MCQ — Factoring

60 Factor $3x^2 + 12x$.

a. $3(x^2 + 4x)$

b. $x(3x + 12)$

c. $3x(x + 12)$

d. $3x(x + 4)$ ✓

GCF = $3x$: $3x^2 + 12x = 3x(x + 4)$.

61 Factor $x^2 - 11x + 30$.

a. $(x - 5)(x - 6)$ ✓

b. $(x - 2)(x - 15)$

c. $(x + 5)(x + 6)$

d. $(x - 3)(x - 10)$

Need two numbers multiplying to 30, adding to -11 : -5 , -6 .

62 Factor $x^2 + 3x - 18$.

a. $(x + 2)(x - 9)$

b. $(x - 6)(x + 3)$

c. $(x + 9)(x - 2)$

d. $(x + 6)(x - 3)$ ✓

Need two numbers multiplying to -18 , adding to 3: 6, -3 .

63 Factor $4x^2 - 25$.

a. $(2x - 25)(2x + 1)$

b. $(2x - 5)(2x + 5)$ ✓

c. $(2x - 5)^2$

d. $(4x - 5)(x + 5)$

Difference of squares: $(2x)^2 - 5^2 = (2x - 5)(2x + 5)$.

MCQ — Solving quadratic equations

64 Solve $x^2 - 9x + 18 = 0$.

- a. $x = -3$ or $x = -6$
- b. $x = 1$ or $x = 18$
- c. $x = 2$ or $x = 9$
- d. **$x = 3$ or $x = 6$ ✓**

$$(x - 3)(x - 6) = 0 \Rightarrow x = 3 \text{ or } x = 6.$$

65 Solve $x^2 + 2x - 24 = 0$.

- a. $x = 6$ or $x = -4$
- b. $x = -2$ or $x = 12$
- c. **$x = -6$ or $x = 4$ ✓**
- d. $x = -3$ or $x = 8$

$$(x + 6)(x - 4) = 0 \Rightarrow x = -6 \text{ or } x = 4.$$

66 The graph shows $y = x^2 - 49$. What are its roots (where the curve crosses the x-axis)?

- a. $x = 7$ only
- b. $x = 49$
- c. **$x = 7$ or $x = -7$ ✓**
- d. $x = -7$ only

The curve crosses the x-axis at $x = -7$ and $x = 7$, matching $x^2 - 49 = 0$.

67 Solve $2x^2 - 8x = 0$.

a. $x = 8$

b. $x = 2$ or $x = 4$

c. $x = 0$ or $x = 4$ ✓

d. $x = 0$ or $x = -4$

$2x(x - 4) = 0 \Rightarrow x = 0$ or $x = 4$.

MCQ — Systems of two equations

68 Solve the system: $x + 2y = 8$, $x - y = 2$.

a. $x = 2$, $y = 3$

b. $x = 6$, $y = 1$

c. $x = 4$, $y = 2$ ✓

d. $x = 8$, $y = 0$

Subtracting: $3y = 6 \Rightarrow y = 2$, then $x = 4$.

69 Solve the system: $3x - y = 7$, $x + y = 5$.

a. $x = 2$, $y = 3$

b. $x = 1$, $y = 4$

c. $x = 3$, $y = 2$ ✓

d. $x = 4$, $y = 1$

Adding the equations: $4x = 12 \Rightarrow x = 3$, then $y = 2$.

70 The graph shows two lines. What is their point of intersection?

- a. (1, 4)
- b. (0, 5)
- c. (2, 3) ✓
- d. (3, 2)

The two lines $y = x + 1$ and $y = -x + 5$ cross where both are true: at (2, 3).

71 Solve the system: $y = 2x - 1$, $y = x + 3$.

- a. $x = -4$, $y = -5$
- b. $x = 2$, $y = 3$
- c. $x = 7$, $y = 4$
- d. $x = 4$, $y = 7$ ✓

$2x - 1 = x + 3 \Rightarrow x = 4$, then $y = 4 + 3 = 7$.

MCQ — Laws of exponents

72 Simplify $b^7 \div b^2$.

- a. $b^{3.5}$
- b. b^{14}
- c. b^9
- d. b^5 ✓

Subtract exponents: $b^{7-2} = b^5$.

73 Simplify $(2x^3)^2$.

a. $4x^6$ ✓

b. $2x^9$

c. $4x^5$

d. $2x^6$

$$(2x^3)^2 = 2^2 x^{3 \times 2} = 4x^6.$$

74 Simplify $x^0 \cdot x^4$ ($x \neq 0$).

a. 1

b. x^5

c. x^0

d. x^4 ✓

$$x^0 = 1, \text{ so } x^0 \cdot x^4 = x^4.$$

75 Simplify m^{-2} .

a. $-\frac{1}{m^2}$

b. $\frac{1}{m^2}$ ✓

c. m^2

d. $-m^2$

$$\text{A negative exponent means reciprocal: } m^{-2} = \frac{1}{m^2}.$$

MCQ — Inequalities

76 The number line shows the solution set of an inequality. Which inequality matches?

- a. $x \geq 4$
- b. $x \leq 4$
- c. $x > 4$
- d. $x < 4$ ✓

An open circle at 4 with shading to the left means $x < 4$.

77 The number line shows the solution set of an inequality. Which inequality matches?

- a. $x \leq -2$
- b. $x < -2$
- c. $x \geq -2$ ✓
- d. $x > -2$

A closed circle at -2 with shading to the right means $x \geq -2$.

78 Solve $-3x + 7 \leq 22$.

- a. $x \geq -5$ ✓
- b. $x \leq -5$
- c. $x \leq 5$
- d. $x \geq 5$

$-3x \leq 15$; dividing by -3 flips the sign: $x \geq -5$.

79 The number line shows the solution set of an inequality. Which inequality matches?

- a. $x > 1$
- b. $x \geq 1$
- c. $x \leq 1$ ✓
- d. $x < 1$

A closed circle at 1 with shading to the left means $x \leq 1$.

MCQ — Reading function graphs

80 The graph shows a line. What is its slope?

a. $\frac{1}{2}$

b. 2 ✓

c. -2

d. -1

The line rises 2 for every 1 across, e.g. from (0, -1) to (1, 1): slope = 2.

81 What is the y-intercept of the line shown?

a. 4 ✓

b. 0

c. -4

d. 1

The line crosses the y-axis at (0, 4).

82 What is the equation of the line shown?

a. $y = 3x + 1$

b. $y = x + 3$

c. $y = \frac{1}{3}x$

d. $y = 3x$ ✓

The line passes through the origin and (1, 3): slope 3, no intercept, so $y = 3x$.

83 Find the slope of the line through the two marked points.

a. 2 ✓

b. $\frac{1}{2}$

c. -2

d. 4

$$\text{Slope} = \frac{6-2}{3-1} = \frac{4}{2} = 2.$$

MCQ — Substitution into formulas

84 Given $A = \frac{1}{2}bh$, find A when $b = 10$ and $h = 6$.

a. 16

b. 8

c. 60

d. 30 ✓

$$A = \frac{1}{2}(10)(6) = 30.$$

85 Given $KE = \frac{1}{2}mv^2$, find KE when $m = 4$ and $v = 3$.

a. 18 ✓

b. 12

c. 24

d. 36

$$KE = \frac{1}{2}(4)(9) = 18.$$

86 The rectangle shown has area $A = lw = 48$ and length $l = 8$. Find the width w .

a. 8

b. 6 ✓

c. 384

d. 40

$$48 = 8w \Rightarrow w = 6.$$

87 Given $F = ma$, find F when $m = 7$ and $a = 5$.

a. 35 ✓

b. 75

c. 12

d. 2.4

$$F = (7)(5) = 35.$$

MCQ — Translating words into equations

88 Three times a number, minus 5, equals twice the number plus 7. Find the number.

a. 19

b. 2

c. -12

d. 12 ✓

$$3n - 5 = 2n + 7 \Rightarrow n = 12.$$

89 The sum of two consecutive even integers is 54. Find the larger integer.

a. 28 ✓

b. 27

c. 29

d. 26

$$n + (n + 2) = 54 \Rightarrow n = 26; \text{ larger integer} = 28.$$

90 Five less than three times a number is 25. Find the number.

a. 10 ✓

b. 20

c. 5

d. 15

$$3n - 5 = 25 \Rightarrow n = 10.$$

91 The perimeter of a rectangle is 54 cm. The length is 3 cm more than the width. Find the width.

a. 24

b. 12 ✓

c. 15

d. 9

$$2(w + (w + 3)) = 54 \Rightarrow 4w + 6 = 54 \Rightarrow w = 12.$$

MCQ — Word problems

92 A number increased by 20% equals 72. Find the number.

- a. 90
- b. 52
- c. 60 ✓
- d. 86.4

$$1.2n = 72 \Rightarrow n = 60.$$

93 Ahmed is twice as old as his sister. In 5 years, the sum of their ages will be 40. Find Ahmed's current age.

- a. 20 ✓
- b. 10
- c. 15
- d. 25

$$\text{Let sister} = x, \text{ Ahmed} = 2x. (x + 5) + (2x + 5) = 40 \Rightarrow 3x = 30 \Rightarrow x = 10; \text{ Ahmed} = 20.$$

94 A positive number plus its square is 30. Find the number.

- a. 30
- b. -6
- c. 6
- d. 5 ✓

$$n^2 + n - 30 = 0 \Rightarrow (n + 6)(n - 5) = 0; \text{ the positive root is } n = 5.$$

95 The bar chart shows four students' test scores. Whose score is exactly 12 more than 70?

- a. Student B
- b. Student D
- c. Student C ✓
- d. Student A

$$70 + 12 = 82, \text{ which is Student C's score.}$$

96 The product of two consecutive integers is 132. Find the larger integer.

a. 13

b. 12 ✓

c. 10

d. 11

$n(n + 1) = 132 \Rightarrow n^2 + n - 132 = 0 \Rightarrow (n + 12)(n - 11) = 0$; taking the positive root $n = 11$, larger
= 12.